HeatGeo: Heat-Diffusion Manifold Distillation for Compact Text Embedding Models

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Abstract

Large text embedding models are central to semantic search, clustering, and retrieval-augmented generation (RAG), where quality depends on the geometry induced by the teacher embedding space. Existing embedding distillation methods provide useful supervision through teacher vectors, token or layer representations, pairwise preferences, and mini-batch relations. These signals, however, do not explicitly describe the intrinsic manifold structure encoded by the teacher's neighborhoods: how local semantic regions connect, how similarity propagates through nearby examples, and which higher-order geometry should remain stable under compression. We propose **HeatGeo**, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo caches teacher embeddings, constructs a mutual k-nearest-neighbor graph as a discrete semantic manifold, and derives multi-scale diffusion distributions that describe how semantic mass flows across this manifold. The student matches these diffusion distributions over sampled candidate neighborhoods, while a spectral manifold objective anchors the student to low-frequency Laplacian coordinates of the teacher graph. HeatGeo therefore turns the teacher's final embeddings into geometry-aware supervision without requiring teacher hidden states, online teacher forward passes, or changes to the student architecture. We evaluate on STS, text classification, and pair classification to study whether manifold diffusion targets improve compact embedding distillation over pointwise, layer-level, and mini-batch relational baselines.

Introduction

Text embedding models have become a basic infrastructure layer for semantic search, clustering, reranking, and retrieval-augmented generation. Their usefulness comes not only from individual vectors, but from the geometric structure those vectors induce: neighborhoods, semantic regions, and smooth paths between related examples. Strong recent embedding models, including LLM-based encoders, often learn rich semantic geometry but are costly to run at scale. Encoding large private collections, refreshing indexes as documents change, or serving retrieval on constrained hardware therefore motivates compact student encoders that inherit the geometry of a stronger teacher.

Knowledge distillation offers a practical route to this compression (Hinton, Vinyals, and Dean 2015). In language representation learning, distillation has been applied through output matching, hidden-state alignment, self-attention transfer, and compact pretrained architectures (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020; Xu et al. 2024). For embedding models, common objectives include pointwise alignment between student and teacher embeddings, contrastive learning with teacher-derived positives, teacher-anchored layer alignment, and relational losses that match pairwise similarities within a mini-batch. These objectives provide valuable supervision at different granularities: individual examples, internal layers, sampled pairs, or local batch structure.

The geometric view suggests a complementary form of supervision. A teacher embedding space can be treated as a sampled semantic manifold: nearby examples define local tangent neighborhoods, multi-hop paths reveal semantic continuity, and low-frequency graph structure captures broader organization. Pointwise losses encourage coordinate agreement, layer-alignment methods improve internal transfer under capacity mismatch, and mini-batch relational losses preserve sampled similarity patterns. HeatGeo asks whether the student can also learn the teacher's intrinsic manifold geometry through diffusion targets derived from the teacher embedding space.

This motivates the central question of this work: can a compact student inherit the teacher's semantic manifold by matching heat-style diffusion on a teacher-induced neighborhood graph? The intuition is simple. If the teacher embeddings define a discrete manifold over training examples, then random walks on this manifold reveal how semantic mass flows from each anchor to nearby examples, broader semantic regions, and higher-order communities. Matching this diffusion process gives the student geometry-aware supervision that is not tied to teacher internals or a particular student architecture.

We propose **HeatGeo**, a heat-diffusion manifold distillation method for compact text embedding models. HeatGeo first caches teacher embeddings and constructs a mutual k-nearest-neighbor graph using teacher cosine similarity. This graph acts as a discrete approximation to the teacher's semantic manifold. From it, HeatGeo defines a transition matrix and precomputes multi-scale diffusion targets corresponding to one-step, two-step, and longer-range semantic random walks. During training, each anchor example is paired with a candidate set containing high-diffusion neighbors, teacher hard negatives, and random negatives. The student then learns a neighborhood distribution over the same candidates and minimizes a weighted KL divergence to the teacher diffusion targets.

HeatGeo adds two stabilizing signals. First, a spectral manifold objective projects student embeddings into low-dimensional coordinates computed from the teacher graph Laplacian, encouraging the student to preserve low-frequency topology of the teacher manifold. Second, a lightweight pointwise anchor aligns projected student embeddings with cached teacher embeddings, acting as a boundary condition rather than the main training signal. A simple curriculum introduces diffusion scales gradually: the student first learns local neighborhoods, then semantic regions, and finally longer-range diffusion with spectral anchoring.

Our contributions are:

- We formulate text embedding distillation as manifold geometry transfer, using the teacher embedding space as a discrete semantic manifold.
- We introduce HeatGeo, which distills multi-scale heat-diffusion distributions from a teacher-induced mutual kNN graph into a compact student encoder.
- We propose a spectral manifold anchoring objective that transfers low-frequency topology from the teacher graph without requiring teacher hidden states or modifying the student architecture.
- We provide an evaluation protocol for HeatGeo across semantic textual similarity, text classification, and pair classification benchmarks, with ablations for diffusion scale, spectral anchoring, candidate sampling, and curriculum.

Related Work

Text Embedding Models and Evaluation. Text embedding models encode variable-length text into fixed-dimensional vectors for retrieval, clustering, semantic textual similarity, and transfer learning. Sentence-BERT made transformer encoders practical for sentence-level similarity search by using siamese and triplet-network training (Reimers and Gurevych 2019), while SimCSE showed that contrastive learning with dropout noise or entailment supervision can produce strong sentence embeddings (Gao, Yao, and Chen 2021). Evaluation suites such as BEIR and MTEB further shifted the field from single-task similarity scores toward broad retrieval, classification, clustering, and semantic textual similarity evaluation (Thakur et al. 2021; Muennighoff et al. 2023). Continued maintenance of MTEB reflects the need for reproducible, multi-task embedding evaluation as models and datasets evolve (Chung et al. 2025).

Recent embedding systems increasingly rely on larger foundation models, richer training mixtures, and specialized retrieval objectives. BGE-M3 combines multilingual, multi-functionality, and multi-granularity training with self-knowledge distillation across dense, sparse, and multi-vector retrieval modes (Chen et al. 2024). Qwen3 Embedding scales embedding and reranking models from Qwen3 foundation models (Zhang et al. 2025), and Dewey targets long-context embedding for RAG-style retrieval (Zhang, Zou, and Zhou 2025). This line of work provides stronger teachers and broader evaluation targets, but it does not by itself specify how a compact student should inherit the intrinsic geometry of the teacher embedding space. HeatGeo treats these models as black-box embedding teachers: it requires only their final embeddings, then distills diffusion geometry induced by their neighborhoods.

Distillation for Language and Embedding Models. Knowledge distillation transfers behavior from a large teacher to a smaller student (Hinton, Vinyals, and Dean 2015). In pretrained language models, DistilBERT, TinyBERT, and MiniLM compress transformer encoders through combinations of logits, hidden states, attention distributions, and self-attention relations (Sanh et al. 2019; Jiao et al. 2020; Wang et al. 2020). Recent LLM distillation extends this toolkit with on-policy training, contrastive objectives, sparse-logit acceleration, and score-matching formulations (Agarwal et al. 2024; Ko et al. 2025; Anshumann et al. 2025; Kim et al. 2025; Xu et al. 2024). Cross-tokenizer methods such as DSKD and CDM address mismatches between teacher and student vocabularies or tokenization schemes (Zhang et al. 2024b; Chen et al. 2025).

Embedding-specific distillation adapts these ideas to retrieval-oriented representations. Gecko distills LLM-generated and LLM-relabelled retrieval data into compact text embedding models (Lee et al. 2024). Jasper and Stella use multi-stage embedding distillation and Matryoshka Representation Learning to improve both quality and dimensional flexibility (Zhang et al. 2024a). LEAF aligns compact embedding students to teacher representations and supports asymmetric retrieval in which queries and documents may be encoded by different models (Vujanic and Rueckstiess 2025). Jina Embeddings v5 combines task-targeted distillation with contrastive training to produce compact multilingual embeddings robust to truncation and quantization (Akram et al. 2026). These methods establish distillation as a central recipe for embedding model development. HeatGeo is complementary to this line: rather than introducing new teacher internals or task labels, it converts cached final embeddings into manifold diffusion targets.

Local and Relational Supervision. Pointwise embedding distillation directly matches a projected student embedding to the corresponding teacher embedding. This is simple and architecture-agnostic, but it asks a compact student to reproduce the teacher's absolute coordinates even when the student's capacity and embedding dimension differ. Contrastive sentence-embedding distillation methods such as DistilCSE improve the training signal by transferring contrastive behavior and pairwise preferences (Gao et al. 2021; Xu et al. 2023). Relational knowledge distillation further argues that students should preserve relations among examples rather than only individual outputs (Park et al. 2019). In embedding training, this idea commonly becomes mini-batch similarity matching: teacher and student pairwise similarity matrices are compared over the examples that happen to co-occur in a stochastic batch.

These local objectives better reflect retrieval than pure coordinate regression, but their supervision is bounded by the sampled batch or pair set. They do not explicitly model multi-hop semantic paths, community structure, or the way probability mass should spread from an anchor across the teacher manifold. Token-level and layer-level alignment methods, including CKA-style representation matching, optimal-transport alignment, EMO-style objectives, and teacher-anchored layer alignment, provide useful intermediate supervision when teacher internals or student layers are available (Kornblith et al. 2019; Alqahtani et al. 2021; Dao et al. 2026). In particular, teacher-anchored layer alignment targets internal transfer and capacity mismatch, while HeatGeo targets the geometry induced by final teacher embeddings. These objectives operate on different aspects of distillation and can be viewed as complementary.

Graph, Manifold, and Diffusion Perspectives. Graph-based manifold learning studies how neighborhood graphs reveal lower-dimensional structure in high-dimensional data. Laplacian Eigenmaps use graph Laplacian eigenvectors to preserve local neighborhoods in low-dimensional representations (Belkin and Niyogi 2003), and Diffusion Maps use Markov diffusion on a graph to expose multi-scale connectivity and geometry (Coifman and Lafon 2006). Knowledge distillation on graphs also studies how structural information can be transferred between graph models or from graph teachers to simpler students (Tian et al. 2025; Zhang et al. 2022). These works show that graph structure and diffusion can encode information unavailable from isolated examples.

HeatGeo draws on this graph-manifold view but applies it to a different distillation setting. The graph is not an observed input graph and the spectral coordinates are not used as a standalone dimensionality-reduction output. Instead, the graph is induced by a strong text embedding teacher, diffusion distributions become soft manifold-neighborhood targets for the student, and Laplacian eigenvectors act as a low-frequency topology anchor. This gives compact text encoders an explicit geometry-aware supervision signal derived from the teacher embedding manifold.

Problem Setup

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote a training corpus. A teacher encoder T maps each input to a teacher embedding

$$t_i = T(x_i) \in \mathbb{R}^{d_T}, \tag{1}$$

and a compact student encoder S_θ maps the same input to

$$s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}. \tag{2}$$

The goal is to train S_θ so that the student embedding space preserves the semantic geometry induced by the teacher, especially nearest-neighbor relations, local semantic regions, and higher-order manifold structure.

Standard pointwise embedding distillation minimizes a distance between each projected student embedding and its teacher embedding:

$$\mathcal{L}_{\text{point}} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \cos(W s_i, t_i)\right), \tag{3}$$

where W is an optional projection. Relational distillation instead matches student and teacher similarity matrices inside a mini-batch B:

$$\mathcal{L}_{\text{rel}} = \left\| G_B^S - G_B^T \right\|_F^2, \tag{4}$$

where G_B^S and G_B^T are cosine-similarity matrices over examples in B. HeatGeo lifts the relational view from mini-batch snapshots to a teacher-induced semantic manifold represented by a neighborhood graph.

HeatGeo

Teacher Semantic Graph

HeatGeo first encodes the full training corpus with the teacher and constructs a teacher semantic graph. For each example x_i, let

$$\mathcal{N}_k^T(i) = \operatorname{TopK}_{j \neq i} \cos(t_i, t_j) \tag{5}$$

be its top-k teacher nearest neighbors. To reduce noisy one-directional edges and hubness, HeatGeo uses a mutual kNN graph:

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j). \tag{6}$$

The weighted adjacency matrix W^T is

$$W_{ij}^T = \begin{cases} \exp(\cos(t_i, t_j)/\tau_g), & j \in \mathcal{M}_k^T(i), \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

where τ_g controls graph sharpness. Let D^T be the diagonal degree matrix. The row-normalized transition matrix is

$$P^T = (D^T)^{-1} W^T. \tag{8}$$

Each row P_i^T defines a one-step semantic random-walk distribution from anchor x_i.

Multi-Scale Heat Diffusion Targets

A one-step transition captures direct neighbors, but embedding spaces also contain higher-order semantic structure. HeatGeo therefore uses multiple diffusion scales:

$$P_r^T = (P^T)^r, \qquad r \in \mathcal{R}, \tag{9}$$

where $\mathcal{R}$ is typically $\{1, 2, 4\}$. Small r preserves sharp local neighborhoods, while larger r captures semantic regions and multi-hop communities. In implementation, diffusion is approximated sparsely by propagating probability mass through graph neighbor lists and retaining top candidates.

For each anchor x_i, HeatGeo builds a candidate set

$$C_i = C_i^+ \cup C_i^h \cup C_i^r, \tag{10}$$

where C_i^+ contains high-probability diffusion neighbors, C_i^h contains teacher hard negatives, and C_i^r contains random negatives. The teacher diffusion target at scale r is the diffusion mass restricted and renormalized over C_i:

$$p_{i,r}^T(j) = \frac{(P_r^T)_{ij}}{\sum_{j' \in C_i} (P_r^T)_{ij'}}, \qquad j \in C_i. \tag{11}$$

This target is a soft distribution over semantic relatedness, rather than a binary positive-negative label.

Student Neighborhood Distribution

Given an anchor embedding s_i and candidate embeddings $\{s_j : j \in C_i\}$, the student induces a distribution over the same candidate set:

$$p_i^S(j) = \frac{\exp(\cos(s_i, s_j)/\tau_s)}{\sum_{j' \in C_i} \exp(\cos(s_i, s_{j'})/\tau_s)}, \qquad j \in C_i, \quad (12)$$

where τ_s is a student temperature. The main diffusion distillation loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}_e} \omega_r D_{\text{KL}} \left(p_{i,r}^T \| p_i^S \right), \qquad (13)$$

where B is a mini-batch, ω_r is the scale weight, and $\mathcal{R}_e$ is the set of active scales at epoch e.

Spectral Manifold Anchoring

Diffusion matching transfers local and multi-hop neighborhoods, but a student can still fit local candidates while distorting broader community structure. HeatGeo therefore computes spectral coordinates from the teacher graph. Let the symmetric normalized graph Laplacian be

$$L^T = I - (D^T)^{-1/2} W^T (D^T)^{-1/2}. \qquad (14)$$

Let $U^T \in \mathbb{R}^{N \times m}$ contain the first m non-trivial eigenvectors of L^T, and let u_i^T be the row corresponding to x_i. A learnable projection maps student embeddings to this spectral space:

$$z_i^S = W_{\text{spec}} s_i. \qquad (15)$$

The spectral objective is

$$\mathcal{L}_{\text{spec}} = \frac{1}{|B|} \sum_{i \in B} \left\| z_i^S - u_i^T \right\|_2^2. \qquad (16)$$

This term encourages the student to retain global teacher topology without requiring hidden states or teacher forward passes during student training.

Training Objective and Curriculum

HeatGeo includes a lightweight pointwise anchor for stability:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|B|} \sum_{i \in B} \left(1 - \cos(W_a s_i, t_i)\right), \qquad (17)$$

where W_a maps student embeddings to the teacher dimension. When paired views are available, we also use a contrastive task loss $\mathcal{L}_{\text{task}}$ implemented as InfoNCE over two student encodings of each example pair.

The final objective is

$$\begin{aligned} \mathcal{L}_{\text{HeatGeo}} = {} & \lambda_{\text{diff}} \mathcal{L}_{\text{diff}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} \\ & + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor}} + \lambda_{\text{task}} \mathcal{L}_{\text{task}}. \end{aligned} \qquad (18)$$

We use a diffusion-scale curriculum: the first epoch uses one-step diffusion, the second epoch adds two-step diffusion, and later epochs add the full scale set and spectral anchoring. This avoids forcing long-range, smoother structure before the student has learned reliable local neighborhoods.

Algorithm 1 HeatGeo Training

Require: Corpus $\mathcal{D}$, teacher T, student S_θ, graph size k, diffusion scales $\mathcal{R}$

Cache teacher embeddings $t_i = T(x_i)$ for all $x_i \in \mathcal{D}$
Build a mutual kNN teacher graph W^T
Compute transition matrix $P^T = (D^T)^{-1} W^T$
Precompute sparse diffusion candidates and targets $p_{i,r}^T$
Compute spectral coordinates u_i^T from the graph Laplacian
for each training epoch **do**
  Activate diffusion scales according to the curriculum
  for each mini-batch B **do**
    Encode anchors and candidate texts with S_θ
    Compute student candidate distributions p_i^S
    Compute $\mathcal{L}_{\text{diff}}$, $\mathcal{L}_{\text{spec}}$, $\mathcal{L}_{\text{anchor}}$, and $\mathcal{L}_{\text{task}}$
    Update θ by minimizing $\mathcal{L}_{\text{HeatGeo}}$
  end for
end for
return Distilled student encoder S_θ

Experiments

Experimental Setup

Models. Our implementation uses a compact BERT/MiniLM-style student encoder and an LLM-based teacher embedding model. Teacher embeddings are cached once before HeatGeo training. After graph construction, the teacher model is removed from GPU memory; HeatGeo training uses cached teacher embeddings, precomputed diffusion targets, and spectral coordinates.

Training Data. We train on text-pair data assembled from multiple semantic tasks. Each training row provides either a text pair or a single text field depending on the task format. HeatGeo constructs the teacher graph over the training examples, using the anchor text for candidate neighborhoods.

Evaluation Tasks and Metrics. We evaluate the distilled student on three task families. For semantic textual similarity, we report Spearman correlation on SICK, STS12, and STS-B. For text classification, we freeze the encoder, train a logistic regression classifier on top of embeddings, and report accuracy and macro-F1 on Banking77, Emotion, and Tweet. For pair classification, we threshold cosine similarity and report accuracy, macro-F1, precision, recall, and average precision on MRPC, SciTail, and WiC.

Baselines. We compare HeatGeo with the undistilled student, pointwise teacher anchoring, mini-batch relational distillation, cross-tokenizer alignment methods such as CDM and DSKD, EMO-style token/attention alignment, Stella-style multi-faceted embedding distillation, TALAS teacher-anchored layer alignment, and the full teacher embedding model when feasible. These baselines cover output-level, layer-level, tokenizer-aware, and relational distillation signals.

Implementation Details. Unless otherwise specified, HeatGeo uses graph size $k = 50$, graph temperature $\tau_g =$

0.1, diffusion scales $\{1, 2, 4\}$, candidate size 64, 32 diffusion positives, 16 hard negatives, 16 random negatives, spectral dimension 16, and five training epochs. The scale weights are $(1.0, 0.5, 0.25)$, giving higher weight to sharper local diffusion. The loss weights are $\lambda_{\text{diff}} = 1.0$, $\lambda_{\text{spec}} = 0.1$, $\lambda_{\text{anchor}} = 0.05$, and $\lambda_{\text{task}} = 0.001$.

Main Results

Table 1 reports downstream embedding quality across the evaluation tasks. The key comparison is whether manifold diffusion supervision improves compact embedding distillation relative to output-level, layer-level, and mini-batch relational baselines while retaining the efficiency of the compact student.

Ablation Study

We isolate the contributions of HeatGeo with the ablations in Table 2. Removing diffusion reduces the method to task loss plus weak anchoring. Removing spectral anchoring tests whether local diffusion alone preserves global topology. Removing the curriculum tests whether long-range diffusion introduced from the start over-smooths early training. Candidate-set ablations test the role of hard and random negatives.

Discussion

HeatGeo is designed around the observation that teacher embeddings define more than isolated targets: they define a semantic manifold whose diffusion geometry can serve as a distillation signal. The method has three practical advantages. First, it uses cached teacher embeddings rather than teacher hidden states, attention maps, or online teacher forward passes, which lowers training-time memory pressure. Second, it is tokenizer-agnostic because the loss is defined over final embeddings and graph diffusion targets. Third, it does not change the student architecture, so the distilled model can be deployed as an ordinary text encoder.

The main limitation is that HeatGeo requires a graph preprocessing step before student training. Building a teacher graph is worthwhile when the same training data supports substantial distillation, but it adds overhead compared with purely mini-batch objectives. HeatGeo also depends on the quality and coverage of the teacher graph: if the teacher embeddings contain spurious neighborhoods, diffusion may propagate those errors. Future work can study approximate nearest-neighbor construction, adaptive diffusion scales, and online graph refresh for changing data.

Conclusion

We introduced HeatGeo, a heat-diffusion manifold distillation framework for compact text embedding models. HeatGeo treats cached teacher embeddings as samples from a semantic manifold, distills multi-scale diffusion over a teacher-induced mutual kNN graph, and anchors low-frequency topology with spectral coordinates. This gives the student a geometry-aware supervision signal while requiring only cached teacher embeddings and no student architecture changes. HeatGeo is therefore a practical manifold-oriented complement to output-level, layer-level, and mini-batch relational embedding distillation objectives.

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Table 1: Experimental results on nine datasets. Datasets marked with ⋆ are in-domain, while the remaining datasets are out-of-domain.

	Classification (F1)			**Pair Classification (AP)**			**STS (Spearman)**			**Domain Avg**		**Avg.**
	Banking77	Tweet	Emotion⋆	MRPC	SciTail	WiC⋆	SICK	STS12	STSB⋆	Avg-In	Avg-Out	Avg.
BGE-M3 → BERT-base												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	–	–	–	–	–	–	–	–	–	–	–	–
DSKD	–	–	–	–	–	–	–	–	–	–	–	–
Jasper and Stella	–	–	–	–	–	–	–	–	–	–	–	–
DistillCSE	–	–	–	–	–	–	–	–	–	–	–	–
EMO	–	–	–	–	–	–	–	–	–	–	–	–
TALAS	–	–	–	–	–	–	–	–	–	–	–	–
HeatGeo	–	–	–	–	–	–	–	–	–	–	–	–
Qwen-Embedding-0.6B → BERT-base												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	–	–	–	–	–	–	–	–	–	–	–	–
DSKD	–	–	–	–	–	–	–	–	–	–	–	–
Jasper and Stella	–	–	–	–	–	–	–	–	–	–	–	–
DistillCSE	–	–	–	–	–	–	–	–	–	–	–	–
EMO	–	–	–	–	–	–	–	–	–	–	–	–
TALAS	–	–	–	–	–	–	–	–	–	–	–	–
HeatGeo	–	–	–	–	–	–	–	–	–	–	–	–
Qwen-Embedding-4B → BERT-base												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	**68.02**	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	**71.64**	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	**73.83**	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	**72.61**	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	92.34	73.74	64.41	**85.97**	82.66	68.65	78.38	72.27	79.98	71.01	80.89	77.60
HeatGeo	**92.62**	72.27	66.42	83.57	**89.15**	66.43	**79.69**	70.43	**81.79**	**71.55**	**81.29**	**78.04**

Table 2: Ablation template for HeatGeo. Report average score across task groups or a selected validation suite.

Variant	Average Score
HeatGeo full	–
w/o diffusion loss	–
w/o spectral anchoring	–
w/o pointwise anchor	–
w/o diffusion curriculum	–
w/o hard negatives	–
w/o random negatives	–
single scale $r = 1$	–
single scale $r = 4$	–

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